Register Number | | | | | | | | | | | | | | | |

PSNA College of Engineering and Technology, Dindigul - 624 622.

(An Autonomous Institution Affiliated to Anna University, Chennai)

End Semester Examinations :: Nov/Dec 2024

B.E	Biomedical Engineering			Max. Marks : 100
Semester:	III	**521193**	**R-2022**	Date : 02.01.2025
BM2312	Electronic Devices and Circuits			Time : 3 Hours FN

Part – A : Answer All Questions. **10 x 2 = 20 Marks** Marks

1	What is the typical forward voltage drop across an LED?	2
2	Define Threshold voltage. Elucidate the factors influencing threshold voltage.	2
3	List the applications of Thyristors.	2
4	State the difference between FINFET and MOSFET.	2
5	Define the current amplification factor.	2
6	Why does a CE amplifier have high input resistance and low output resistance?	2
7	How do neutralization methods counteract the effects of positive feedback in amplifiers?	2
8	Summarize the advantages and disadvantages of using a single-tuned amplifier compared to other amplifiers.	2
9	What is negative feedback? List the advantages of negative feedback in amplifiers.	2
10	Outline the significance of the gain-phase margin and phase margin in feedback amplifiers.	2

Part - B : Answer All Questions. **5 x 13 = 65 Marks**

11	A	Explain the working and V-I characteristics of the PN junction diode and Zener diode.	13
		OR	
	B	Describe the working of a full wave rectifier and derive its average voltage, RMS voltage, Peak factor, Ripple factor and efficiency.	13
12	A	Explain the construction and operation of an NPN transistor with a neat sketch. Also comment on the characteristics of the NPN transistor.	13
		OR	
	B	(i) Describe the construction and working of UJT with its equivalent circuit and V-I characteristics.	10
		(ii) In a transistor, I_B = 68 μA, I_E = 30 mA and β = 440. Determine the α rating of the transistor and the value of I_C using both the α rating and β rating of the transistor.	3
13	A	An amplifier rated at 38W output is connected to a 10Ω speaker.	7
		(i) Calculate the input power required for full power output if the power gain is 20 dB.	
		(ii) Calculate the input voltage for rate output if the amplifier voltage gain is 40 dB.	6
		OR	

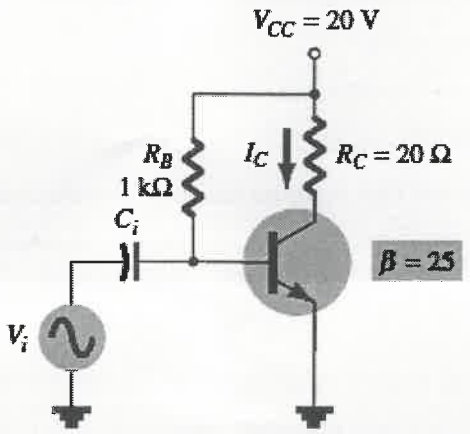
B Explain the small signal analysis of a common emitter amplifier and derive the necessary equations to calculate the voltage gain, input and output impedance. 13

14 A Draw the circuit diagram for an emitter-coupled differential amplifier using BJTs. Describe the common mode and differential mode of working. 13

OR

B (i) What is cross-over distortion? How does it occur in power amplifiers? 5

(ii) Calculate the input power, output power and efficiency of the amplifier circuit shown in Figure 1 for an input voltage that results in a base current of 10 mA peak. 8



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Figure 1

15 A The current series feedback types of transistor amplifiers have the following parameters: R_1=20kΩ, R_2=20kΩ, h_{ie}=2kΩ, R_L=1kΩ, R_e=100 Ω, h_{fe}=80. Calculate A, β, R_{if}, A_f and the loop gain in dB. 13

OR

B (i) What are the necessary conditions for oscillation? State the role of thermal noise in oscillators. 5

(ii) Explain the operation of the Colpitts oscillator with a neat circuit diagram and waveform. 8

Part – C : Answer All Questions. 1 x 15 = 15 Marks

16 A (i) When two PN diodes are connected in series with a voltage source of 1.2V, what is the potential drop? How much current flows in the circuit? 5

(ii) Design a Hartley oscillator for the frequency of 200 kHz and draw the circuit diagram. 10

OR

16 B Determine the Q point of the transistor circuit shown in Figure 2. Also draw the d.c. load line. 15
Given $\beta = 200$ and $V_{BE} = 0.7V$.

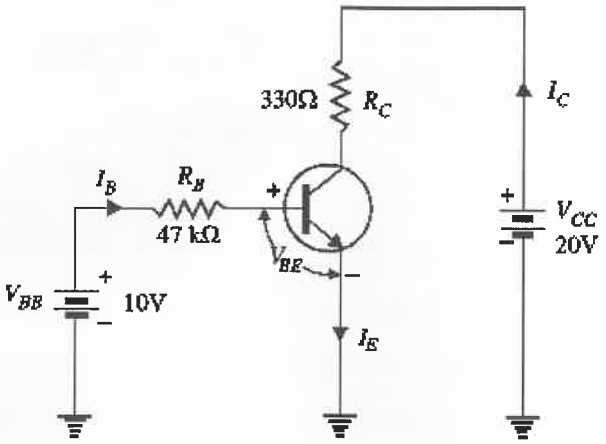


Figure 2

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